

L.Tech. Degree VI Semester Examination April 2014**CS/IT 603 OPERATING SYSTEMS**
(2006 Scheme)

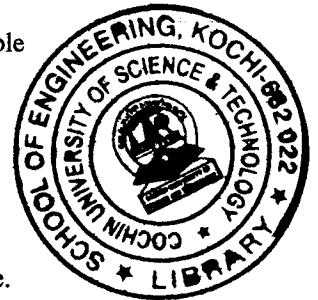
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) What are semaphores? What are the operations associated with semaphore?
- (b) What do you mean by race condition? Give an example of how race condition occurs in a system.
- (c) What is the purpose of a page table? Give the structure of a typical page table entry.
- (d) Explain Least Recently used page replacement algorithm.
- (e) Explain how files are structured.
- (f) Explain device drivers in I/O software.
- (g) Define a dead lock. What are the necessary conditions to occur a dead lock?
- (h) Explain the concept of safe and unsafe state in the content of deadlock avoidance.

**PART B**

(4 x 15 = 60)

- II. (a) Explain with algorithm how monitors can be used to achieve mutual exclusion. (10)
 - (b) How is two level scheduling is done? (5)
- OR**
- III. (a) Describe how the producer consumer problem can be solved with message passing. (10)
 - (b) Explain priority scheduling algorithm (5)
- IV. (a) Differentiate between multiprogramming with fixed partitions and variable partitions. (7)
 - (b) If FIFO page replacement is used with 4 page frames and eight pages, how many page faults will occur with reference string 7103017232, if the 4 frames are initially empty? Repeat the same reference string for LRU also and find the number of page faults. (8)
- OR**
- V. (a) Write short notes on (15)
 - (i) Associative memory
 - (ii) Inverted page table
 - (iii) Memory management using linked lists.
- VI. (a) Explain different disk scheduling algorithms in detail with suitable example. (15)
- OR**
- VII. (a) What are the functions of device controllers? (5)
 - (b) Explain various file allocation methods. (10)
- VIII. (a) What are the ways to recover from a dead lock? (5)
 - (b) Describe Bankers' algorithm for deadlock avoidance with multiple resources of each type. (10)
- OR**
- IX. (a) How can dead lock be prevented in a system? (8)
 - (b) Explain the deadlock detection algorithm for single resource of each type. (7)